A. Vowels and consonants

Albert does not agree with the basic English alphabet. He thinks, that in the alphabet, first, we need to print vowel letters first and then consonants, and both vowels and consonants are ordered alphabetically. So he asks us to sort string according to his own alphabet.

Input

The first line contains one integer n ($1 \le n \le 10^5$) — the length of the string.

The second line contains the string s of exactly n lowercase Latin letters.

Output

Print the letters of s rearranged according to Albert's alphabet: first all its vowels in alphabetical order, then all its consonants in alphabetical order.

Note

Vowel letters are a, e, i, o and u.

Examples

Input 1

```
41
hqghumeaylnlfdxfircvscxggbwkfnqduxwfnfozv
```

Output 1

```
aeiouubccddfffffgggghhkllmnnnqqrsvvwwxxxyz
```

B. House of love

Once upon a time, in one university, created a one organization called "House of love" in which friendship is the main value. So they created one test to know level of their friendship. Two person write some numbers and gives it to you, to check common numbers.

Input

The first line contains two integers n and m ($0 \le n, m \le 10^5$) — the sizes of the two arrays.

The second line contains n integers — the numbers of the first person, and the third line contains m integers — the numbers of the second person ($|a_i| \le 10^9$). If an array is empty, the corresponding line is empty or absent.

Output

Print the numbers that occur in both arrays, **in non-decreasing order**, separated by spaces.

A number that occurs x times in the first array and y times in the second one must be printed exactly $\min(x, y)$ times. If there are no common numbers, print an empty line.

Examples

Input 1

```
4 2
1 2 2 1
2 2
```

Output 1

```
2 2
```

Input 2

```
3 5
4 9 5
4 3 2 1 9
```

Output 2

```
4 9
```

Input 3

```
0 1
1
```

Output 3

```
```

C. Points in Proximity

You are given a list of integer points in a line. Find the pair of points with the least absolute difference. If there are more than one pairs output them all.

Input

The first line contains an integer $N (2 \leq N \leq 2 \times 10^5)$, number of points. The next line represents N integer numbers $(-10^7 \leq points[i] \leq 10^7)$ denoting the points in a line.

Output

Sort the points. Print all pairs of **neighbouring** points of the sorted sequence whose difference is the smallest one, in non-decreasing order: for every such pair print its two points one after another. All numbers are printed in one line, separated by single spaces.

If several points are equal, each neighbouring pair of them is printed separately.

Examples

Input 1

```
6
-20 -3916237 -357920 -362060 30 6246457
```

Output 1

```
-20 30
```

D. Calendar

You're given n dates in format $DD - MM - YYYY$. Sort them in ascending and print in te same format.

Input

The first line contains one integer n $(1 \leq n \leq 10^3)$. Next n lines contain dates in format $DD - MM - YYYY$ $(1 \leq D \leq 31, 1 \leq M \leq 12, 1900 \leq Y \leq 3000)$. Every date is a valid calendar date.

Output

Print sorted dates in the same format as the input.

Examples

Input 1

```
3
01-12-2000
01-11-2000
31-10-2000
```

Output 1

```
31-10-2000
01-11-2000
01-12-2000
```

Input 2

```
4
01-12-1999
12-11-2002
20-02-1900
31-01-1999
```

Output 2

```
20-02-1900
31-01-1999
01-12-1999
12-11-2002
```

E. Incomplete sorting

You are given matrix with n rows and m columns.

You have to sort each column of matrix in descending order.

Input

The first line consists of two integers n and m ($1 \le n, m \le 100$) — the number of rows and columns of the matrix respectively.

Each of the next n lines contains m integers $d_{i,j}$ ($-10^9 \le d_{i,j} \le 10^9$) — elements of the matrix at the intersection of the i-th row and the j-th column.

Output

Print this matrix after sorting.

Examples

Input 1

```
4 3
1 2 3
4 5 6
6 7 8
9 8 7
```

Output 1

```
9 8 8
6 7 7
4 5 6
1 2 3
```

F. New GPA

KBTU introduces new GPA calculation system. You're given n students, their marks and number of credits for their subjects. Sort them by total GPA. It is calculated as $\frac{\sum_{i=1}^{m} g_i * c_i}{\sum_{i=1}^{m} c_i}$, where m - student's number of subjects, g_i, c_i - GPA and number of credits for i-th subject respectively. GPA scale is given in the notes.

Input

First line contains one integer n $(1 \le n \le 10^5)$. Each of the next n lines contains information about the i-th student: his lastname, firstname, an integer m_i $(1 \le m_i \le 10)$ - number of subjects for this student, then m_i marks and number of credits for each subject.

Output

You should print sorted list of students. Each student should be printed in the following format: $lastname\ firstname\ GPA$. First, sort them by overall GPA **in ascending order**; if the GPA is equal, sort by lastname, then by firstname. GPA should be printed with three digits after decimal point.

The GPA must be printed exactly as `printf("%.3f", gpa)` prints it in C++ (equivalently, `cout « fixed « setprecision(3) « gpa`). This matters: for many students the GPA falls exactly on the rounding boundary, and other rounding rules give a different answer.

Note

A+	**4.00**
A	3.75
B+	3.50
B	3.00
C+	2.50
C	2.00
D+	1.50
D	1.00
F	0

Examples

Input 1

```
5
Issenbayev Yernur 4 A 4 D+ 2 B 3 A+ 4
Yermekbayeva Diana 3 A+ 4 B+ 3 B 1
Kadyrov Asman 2 A+ 4 A+ 4
Stepanenko Ivan 3 C+ 3 F 1 A+ 5
Bissimbayev Arystan 3 A+ 4 A+ 5 D 1
```

Output 1

```
Stepanenko Ivan 3.056
Issenbayev Yernur 3.308
Yermekbayeva Diana 3.688
Bissimbayev Arystan 3.700
Kadyrov Asman 4.000
```

G. Nurbol hacker

This is a continuation of the story about the Aslan and Password task from the last quiz. The story is that Nurbol successfully hacked Aslan's Steam account. Aslan was of course upset, but he doesn't give up. He decided to restore his Steam account. To do this, he wrote in support of Steam. Aslan gave them the nickname of his account. But it is possible for Nurbol to change the nickname of the account. And now Steam employees must understand what the new nickname is. Employees have logs of changing nicknames, the logs consist of n lines. The line itself consists of two words, the old nickname, and the new one. Help employees to display all original nicknames and new nicknames next to them.

Input

The first line is number n ($1 \leq n \leq 1000$) the number of requests to change the nickname. Next n lines consist of two strings, old nickname and new nickname.

Every nickname is a non-empty string of at most 20 Latin letters and digits; nicknames are case-sensitive, and the old nickname always differs from the new one.

It is guaranteed that a nickname belongs to one user forever: once some user has used a nickname, no other user can ever take it.

Output

The first line is the number q — the number of users who changed their nicknames.

Each of the next q lines contains two strings: the original nickname of a user and his current nickname. Print these lines **sorted by the original nickname in lexicographical order**.

Note

In the first test nickname "Aslan" was changed to nickname "Nurbol". Then this nickname was changed to "HackMachine". As a result, the original nickname "Aslan" became "HackMachine".

Examples

Input 1

```
2
Aslan Nurbol
Nurbol HackMachine
```

Output 1

```
1
Aslan HackMachine
```

Input 2

```
6
Sens3i Danya
S1mple Papa
M9snoyPovar AWPMaster
IAmNoob IAmPro
Papa Sanya
IAmPro IAmNoob
```

Output 2

```
4
IAmNoob IAmNoob
M9snoyPovar AWPMaster
S1mple Sanya
Sens3i Danya
```

H. Problem 8

You have one letter. Your task is to find "balanced" char in array. "Balanced" char is the smallest in array, but more than your letter.

Input

The first line contains one integer n ($2 \le n \le 2 \cdot 10^5$) — the number of letters in the array.

The second line contains n lowercase Latin letters separated by single spaces, given in non-decreasing order.

The third line contains one lowercase Latin letter k.

Output

Print the “balanced” letter — the smallest letter of the array that is greater than k.

If the array contains no letter greater than k, the search wraps around and the answer is the first (that is, the smallest) letter of the array.

Note

Size of array should be more than 1;

Examples

Input 1

```
3
c f g
a
```

Output 1

```
c
```

I. Problem C. Azat likes sorting

Azat is a perfectionist and he likes everything to be perfect. Help Azat sort letters alphabetically. good luck!

Input

The only line contains the string s of lowercase Latin letters ($1 \le |s| \le 10^6$).

Output

Print the letters of s in non-decreasing alphabetical order.

Note

Try to sort the letters yourself instead of calling the library sort — at the defence you will be asked to do exactly that.

Examples

Input 1	Output 1
qwertyuioplkjvbn	beijklnopqrtuvwy